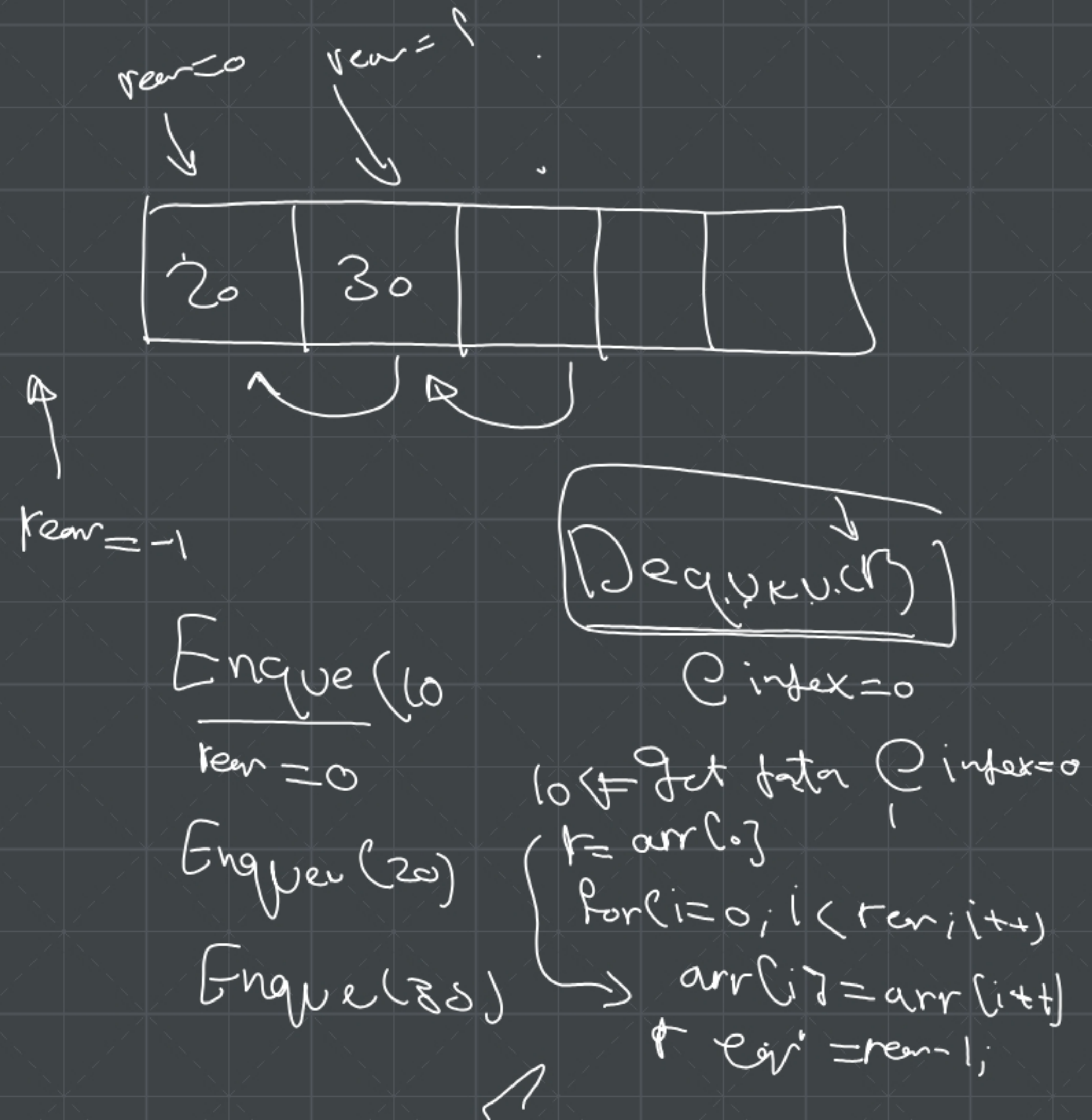


Queue
→ FIFO

Enque $O(1)$

Deque $O(n)$




```

class Stack
{
    Node* tos;
    public
    Stack()
    {
        tos = NULL;
    }
    isEmpty()
    {
        return tos == NULL;
    }
    isFull()
    {
        Node* temp = new Node();
        if (temp == NULL)
            return 1;
        delete temp;
        return 0;
    }
}

```

```

struct node
{
    int d;
    node* prev;
}

```

```

Push (int d)
{

```

```

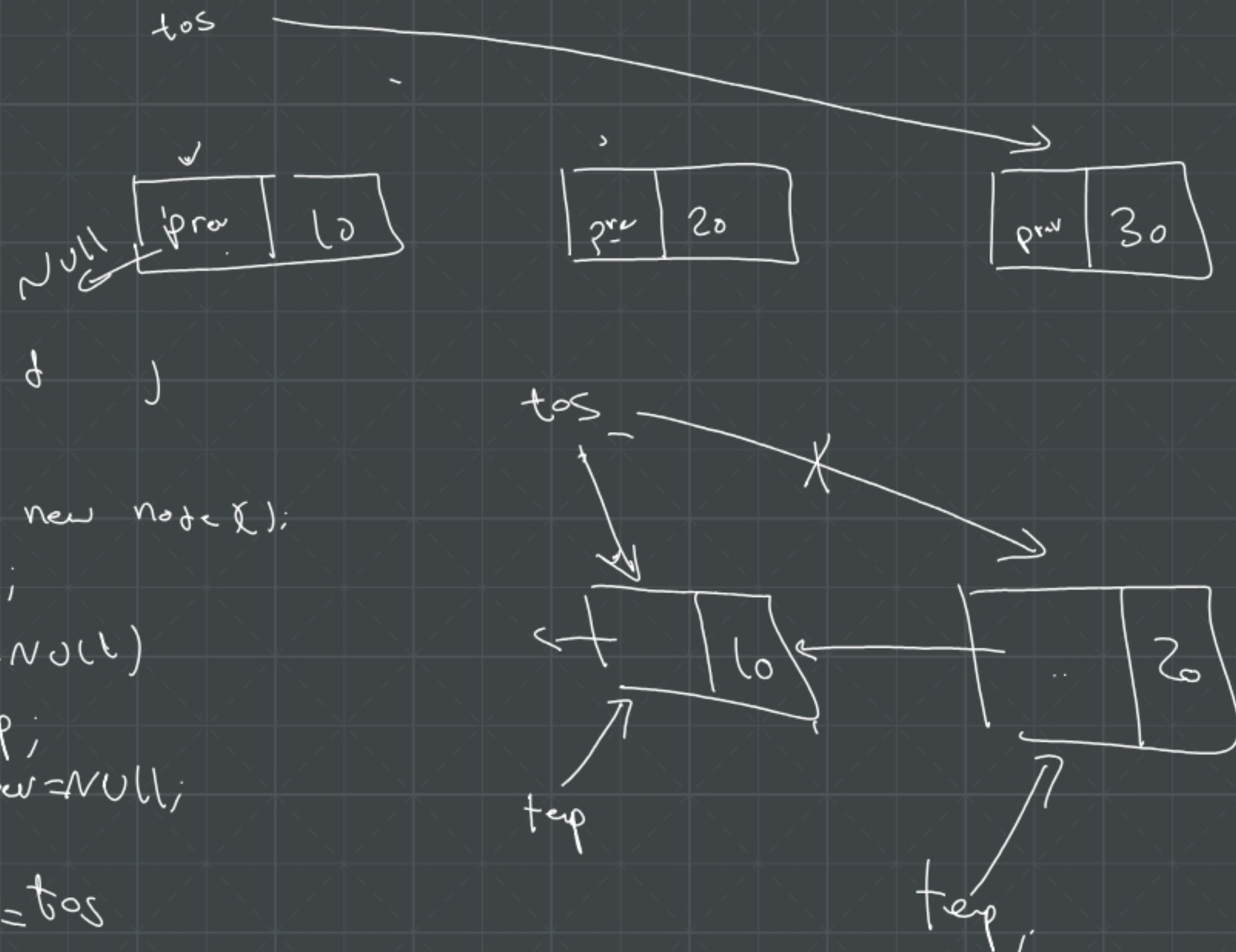
    Node* temp = new Node();
    temp->d = d;
    if (tos == NULL)
    {
        tos = temp;
        tos->prev = NULL;
    }
    else
    {
        temp->prev = tos;
        tos = temp;
    }
}

```

```

int pop (int d)
{
    if (tos == NULL)
        return -1;
    d = tos->d;
    Node* temp = tos;
    tos = tos->prev;
    delete temp;
}

```



Queue

```

{
    node* front,
    node* tail;
}

```

ctor

```

{
    front = rear = NULL;
}

```

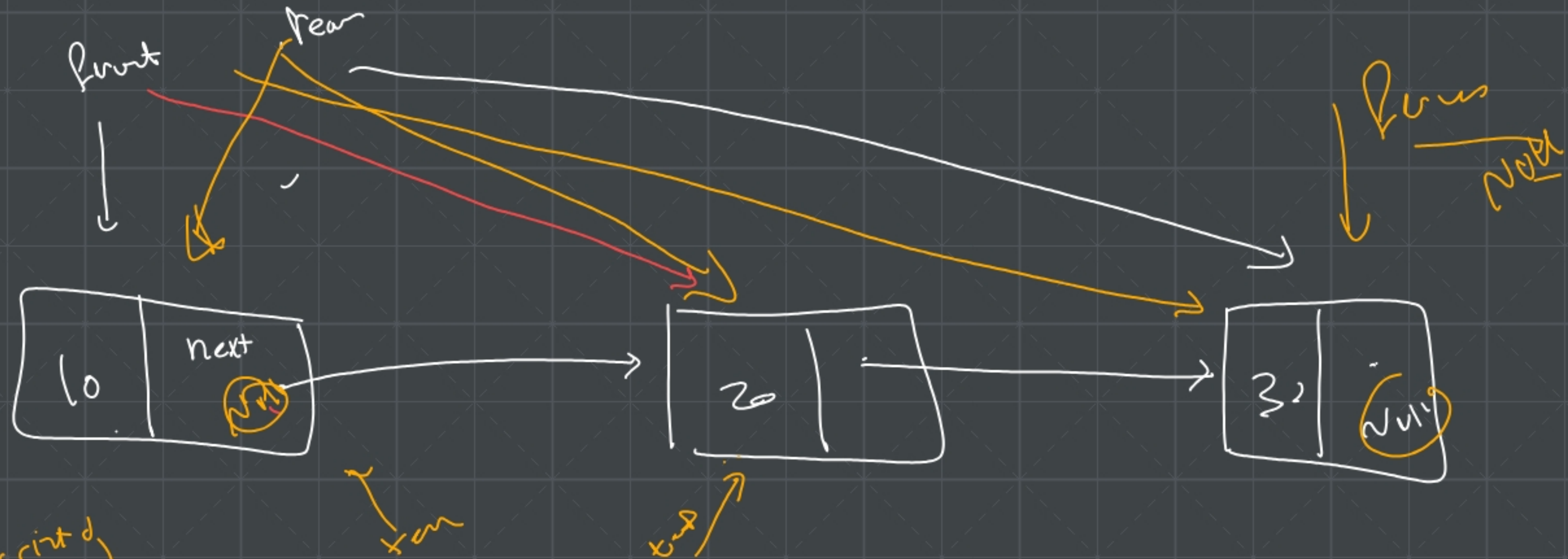
😊

}

```

node {
    int d;
    node* next;
}

```



Enqueue (int d)

```

node* temp = new node();
temp->d = d;
temp->next = NULL;
if (front == NULL)
    front = rear = temp;
else
    rear->next = temp;
    rear = temp;
}

```

```

int Dequeue (int &d)
{
    if (front == NULL)
        return -1;
    else
        d = front->d;
        node* temp = front;
        front = front->next;
        delete temp;
        if (front == NULL)
            rear = NULL;
}

```

😊

